1 **(a)** $7^2 = 5^2 + x^2 - 2(5)(x)\cos 60°$ **[M1]**

$\cos 60° = \frac{1}{2}$ or 0.5 seen **[B1]**

$49 = 25 + x^2 - 5x \implies x^2 - 5x - 24 = 0$ (with no errors seen) **[A1]**

(b) $(x - 8)(x + 3) = 0$ **[M1]**

$x = 8$ **[A1]**

(c) Area $= \frac{1}{2} \times 5 \times 8 \times \sin 60°$ **[M1]**

$10\sqrt{3}$ **[A1]**

[Total Marks: 7]

2 **(a)** $0.5 \times 8 \times 12 \times \sin(42°)$ **[M1]**

32.1 (or 32.11 to 32.12) **[A1]**

(b) $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos(42°)$ **[M1]**

$BC^2 = 65.3...$ **[M1]**

8.08 (or 8.081 to 8.082) **[A1]**

[Total Marks: 5]

3 **(a)** $\frac{1}{2} \times 8 \times 12 \times \sin 40°$ **[M1]**

30.9 (accept $30.85\ldots$) **[A1]**

(b) $8^2 + 12^2 - 2 \times 8 \times 12 \times \cos 40°$ **[M1]**

$60.9\ldots$ **[A1]**

7.80 or 7.81 **[A1]**

[Total Marks: 5]

4 **(a)** $0.5 \times 8.5 \times 11.2 \times \sin(54°)$ **[M1]**

38.5 (38.50 to 38.51) **[A1]**

(b) $BC^2 = 8.5^2 + 11.2^2 - 2(8.5)(11.2)\cos(54°)$ [M2]
(or [M1] for correct cosine rule formula with one substitution error)

$BC^2 = 85.77...$ **[A1]**

9.26 (9.261...) **[A1]**

[Total Marks: 5]

5 **(a)** $\frac{1}{2} \times 8 \times 12 \times \sin(50°)$ **[M1]**

36.77... so 36.8 **[A1]**

(b) $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos(50°)$ **[M1]**

$BC^2 = 208 - 123.415... = 84.58...$ **[M1]**

9.197... so 9.20 **[A1]**

[Total Marks: 5]

6 (a) $\frac{1}{2} \times 7.2 \times 9.5 \times \sin 64°$ **[M1]**

30.7 or 30.73... **[A1]**

(b) $7.2^2 + 9.5^2 - 2(7.2)(9.5)\cos 64°$ **[M2]**

(or $M1$ for correct implicit cosine rule equation)

9.06 or 9.062... **[A1]**

[Total Marks: 5]

7 (a) Uses cosine rule: $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos 65°$ **[M1]**

$BC^2 = 126.85...$ **[M1]**

11.3 **[A1]**

(b) Uses area formula: $\frac{1}{2} \times 8 \times 12 \times \sin 65°$ **[M1]**

43.5 **[A1]**

[Total Marks: 5]

8 (a) $\frac{1}{2} \times 8 \times 11 \times \sin(65°)$ **[M1]**

39.9 **[A1]**

(b) $BC^2 = 8^2 + 11^2 - 2(8)(11)\cos(65°)$ **[M2]**

($M1$ for correct substitution into implicit cosine rule)

10.5 **[A1]**

[Total Marks: 5]

9 (a) $\frac{BC}{\sin 45°} = \frac{12}{\sin 60°}$ **[M1]**

$BC = \frac{12 \sin 45°}{\sin 60°}$ **[M1]**

9.80 **[A1]**

(b) Angle $ACB = 180° - (45° + 60°) = 75°$ **[B1]**

Area $= \frac{1}{2} \times 12 \times 9.798 \times \sin 75°$ **[M1]**

56.8 **[A1]**

[Total Marks: 6]

10 **(a)** $\dfrac{BC}{\sin 40^\circ} = \dfrac{12}{\sin 60^\circ}$ **[M1]**

$BC = \dfrac{12 \sin 40^\circ}{\sin 60^\circ}$ **[M1]**

8.91 (accept 8.907 . . .) **[A1]**

(b) Angle $BCA = 180^\circ - (40^\circ + 60^\circ) = 80^\circ$ **[B1]**

$\dfrac{1}{2} \times 12 \times 8.91 \times \sin 80^\circ$ **[M1]**

52.6 (accept 52.62 . . . to 52.65 . . .) **[A1]**

[Total Marks: 6]

11 **(a)** $\cos Q = \dfrac{7^2 + 8^2 - 12^2}{2 \times 7 \times 8}$ [M2] (M1 for correct implicit cosine rule equation e.g. $12^2 = 7^2 + 8^2 - 2(7)(8)\cos Q$)

106.1 (accept 106.07 . . .) **[A1]**

(b) $\dfrac{1}{2} \times 7 \times 8 \times \sin 106.1^\circ$ **[M1]**

26.9 (accept 26.89 . . .) **[A1]**

[Total Marks: 5]

12 **(a)** $\dfrac{\sin P}{12} = \dfrac{\sin 35^\circ}{8}$ **[M1]**

$\sin P = \dfrac{12 \sin 35^\circ}{8} = 0.8603...$ **[M1]**

Acute angle $= 59.35...^\circ$ **[A1]**

Obtuse angle $P = 180^\circ - 59.35...^\circ = 120.64...^\circ$ so 120.6° **[A1]**

(b) Angle $R = 180^\circ - 35^\circ - 120.64...^\circ = 24.36...^\circ$ **[M1]**

Area $= \dfrac{1}{2} \times 8 \times 12 \times \sin(24.36...^\circ) = 19.79...$ so 19.8 **[A1]**

[Total Marks: 6]

13 **(a)** $\dfrac{\sin R}{15} = \dfrac{\sin 42^\circ}{12}$ **[M1]**

$\sin R = \dfrac{15 \sin 42^\circ}{12}$ (evaluates to 0.836 . . .) **[M1]**

123.2 (accept 123.2 to 123.3) **[A1]**

(b) Angle $P = 180^\circ - 42^\circ - 123.24^\circ = 14.76^\circ \ldots$ **[M1]**

$\dfrac{QR}{\sin 14.76^\circ} = \dfrac{12}{\sin 42^\circ}$ (or equivalent application of sine/cosine rule) **[M1]**

4.57 (accept 4.56 to 4.58) **[A1]**

[Total Marks: 6]

14 **(a)** $\dfrac{\sin R}{14} = \dfrac{\sin 35^\circ}{10}$ **[M1]**

$\sin^{-1}\left(\dfrac{14\sin 35^\circ}{10}\right)$ leading to 53.4° (53.41...) **[M1]**

126.6 (126.58...) **[A1]**

(b) Angle $PQR = 180^\circ - 35^\circ - 126.58^\circ = 18.4^\circ$ **[M1]**

Area $= \dfrac{1}{2} \times 14 \times 10 \times \sin(18.4^\circ)$ **[M1]**

22.1 (22.11...) **[A1]**

[Total Marks: 6]

15 **(a)** $\dfrac{\sin Q}{14} = \dfrac{\sin 32^\circ}{9}$ **[M1]**

$\sin Q = \dfrac{14\sin 32^\circ}{9}$ or 0.824... **[M1]**

Acute angle $= 55.5^\circ$ (55.51...) **[A1]**

$180 - 55.5... = 124.5^\circ$ (124.48 to 124.5) **[A1]**

(b) Angle $PRQ = 180 - 32 - 124.48 = 23.52^\circ$ **[B1]**

$0.5 \times 14 \times 9 \times \sin(23.52^\circ)$ **[M1]**

25.1 (25.12 to 25.14) **[A1]**

[Total Marks: 7]

16 **(a)** $\cos Q = \dfrac{15^2 + 12^2 - 18^2}{2 \times 15 \times 12}$ **[M1]**

$\cos Q = \dfrac{45}{360}$ OR 0.125 **[M1]**

$Q = \cos^{-1}(0.125) = 82.819...^\circ$ **[A1]**

(b) $0.5 \times 15 \times 12 \times \sin(82.8^\circ)$ **[M1]**

89.3 (or 89.28 to 89.30) **[A1]**

(c) Area $= 0.5 \times 15 \times h = 89.3$ OR $h = 12 \times \sin(82.8^\circ)$ **[M1]**

11.9 (or 11.90 to 11.91) **[A1]**

[Total Marks: 7]

17 **(a)** $\dfrac{\sin C}{12} = \dfrac{\sin 35^\circ}{9}$ **[M1]**

$\sin C = \dfrac{12\sin 35^\circ}{9}$ $(= 0.764...)$ **[M1]**

49.9° (or 49.88...) **[A1]**

130.1° (or 130.11...) **[A1]**

(b) Angle $ABC = 180 - (35 + 130.1) = 14.9^\circ$ **[M1]**

Area $= \frac{1}{2} \times 12 \times 9 \times \sin(14.9°)$ **[M1]**

13.9 (accept 13.87... to 13.89...) **[A1]**

[Total Marks: 7]

18 **(a)** Uses sine rule: $\frac{\sin R}{15.2} = \frac{\sin 72°}{18.5}$ **[M1]**

$\sin R = 0.781...$ **[M1]**

51.4 **[A1]**

(b) Angle $P = 180° - 72° - 51.38° = 56.62°$ **[B1]**

$\frac{QR}{\sin 56.62°} = \frac{18.5}{\sin 72°}$ **[M1]**

16.2 **[A1]**

(c) Shortest distance $= 15.2 \times \sin 72°$ **[M1]**

14.5 **[A1]**

[Total Marks: 8]

19 **(a)** $BD^2 = 8^2 + 10^2 - 2(8)(10)\cos 110°$ **[M1]**

$BD^2 = 164 - 160(-0.342) = 218.72$ **[M1]**

14.8 **[A1]**

(b) $\cos C = \frac{14^2 + 15^2 - 14.79^2}{2(14)(15)}$ **[M1]**

$\cos C = \frac{202.26}{420} = 0.4816$ **[M1]**

61.2 **[A1]**

(c) Area $ABD = \frac{1}{2}(8)(10)\sin 110° = 37.59$ **[M1]**

Area $BCD = \frac{1}{2}(14)(15)\sin 61.2° = 92.01$ **[M1]**

130 (or 129.6) **[A1]**

[Total Marks: 9]

20 **(a)** $\frac{\sin C}{10.5} = \frac{\sin 35°}{8.2}$ **[M1]**

$\sin C = \frac{10.5 \times \sin 35°}{8.2}$ OR 0.734... **[M1]**

$C = 180° - 47.3° = 132.7°$ (or 132.73 to 132.74) **[A1]**

(b) Angle $B = 180° - 35° - 132.7° = 12.3°$ **[M1]**

$\frac{AC}{\sin 12.3°} = \frac{8.2}{\sin 35°}$ OR $AC^2 = 10.5^2 + 8.2^2 - 2(10.5)(8.2)\cos 12.3°$ **[M1]**

3.04 (or 3.035 to 3.045) **[A1]**

(c) $0.5 \times 3.04 \times 6.4 \times \sin(ACD) = 8.5$ **[M1]**

$\sin(ACD) = \dfrac{8.5}{9.728}$ OR Angle $ACD = \sin^{-1}\left(\dfrac{8.5}{9.728}\right) = 60.9°$ (or 60.8... to 61.1°) **[A1]**

[Total Marks: 8]

21 (a) $\dfrac{\sin F}{10.5} = \dfrac{\sin 42°}{8.2}$ **[M1]**

$\sin F = 0.8568\ldots$ yielding acute angle $58.96\ldots°$ **[A1]**

$180° - 58.96° = 121.0$ (accept $121.04\ldots$) **[A1]**

(b) Angle $EDF = 180° - 42° - 121.04° = 16.96°$ **[B1]**

$\dfrac{EF}{\sin 16.96°} = \dfrac{8.2}{\sin 42°}$ **[M1]**

3.57 (accept $3.56\ldots$ to $3.58\ldots$) **[A1]**

(c) Shortest distance $= 10.5 \times \sin 42°$ **[M1]**

7.03 (accept $7.025\ldots$ to $7.027\ldots$) **[A1]**

[Total Marks: 8]

22 (a) $BD^2 = 6^2 + 8^2 - 2(6)(8)\cos(110°)$ **[M1]**

$BD^2 = 100 - 96(-0.342...) = 132.83...$ **[M1]**

$BD = 11.52...$ so 11.5 **[A1]**

(b) $\cos C = \dfrac{9^2 + 5^2 - (11.525...)^2}{2(9)(5)}$ **[M1]**

$\cos C = \dfrac{106 - 132.83...}{90} = -0.2981...$ **[M1]**

$C = 107.34...°$ so 107.3° **[A1]**

(c) Area of triangle $ABD = \dfrac{1}{2} \times 6 \times 8 \times \sin(110°) = 22.55...$ **[M1]**

Area of triangle $BCD = \dfrac{1}{2} \times 9 \times 5 \times \sin(107.34...°) = 21.47...$ **[M1]**

Total Area $= 22.55... + 21.47... = 44.02...$ so 44.0 **[A1]**

[Total Marks: 9]

23 (a) $7^2 + 9^2 - 2 \times 7 \times 9 \times \cos 110°$ **[M1]**

$173.09\ldots$ **[A1]**

13.2 (accept 13.15 to 13.2) **[A1]**

(b) $13.156^2 = 10^2 + 14^2 - 2 \times 10 \times 14 \times \cos(\angle BCD)$ **[M1]**

$\cos(\angle BCD) = \dfrac{10^2 + 14^2 - 173.09}{2 \times 10 \times 14}$ **[M1]**

64.0 (accept $63.9\ldots$ to 64.0) **[A1]**

(c) $\frac{1}{2} \times 7 \times 9 \times \sin 110°$ (evaluates to 29.6 . . .) **[M1]**

$\frac{1}{2} \times 10 \times 14 \times \sin 64.0°$ (evaluates to 62.9 . . .) **[M1]**

92.5 (accept 92.4 to 92.6) **[A1]**

[Total Marks: 9]

24 (a) $AC^2 = 7.4^2 + 9.1^2 - 2(7.4)(9.1)\cos(112°)$ [M2]
(or [M1] for correct cosine rule formula with one error)

$AC = 13.7$ (13.71 to 13.72) **[A1]**

(b) $\frac{\sin D}{13.712} = \frac{\sin 43°}{10.5}$ **[M1]**

$\sin D = \frac{13.712 \sin 43°}{10.5}$ or 0.890... **[M1]**

63.0 (62.95 to 63.0) **[A1]**

(c) Angle $ACD = 180 - 43 - 62.95 = 74.05°$ **[M1]**

Shortest distance $= 10.5 \times \sin(74.05°)$ **[M1]**

10.1 (10.09 to 10.1) **[A1]**

[Total Marks: 9]

25 (a) $BD^2 = 7.5^2 + 9.2^2 - 2(7.5)(9.2)\cos(105°)$ **[M2]**

($M1$ for correct substitution into implicit cosine rule)

13.3 (13.28...) **[A1]**

(b) $\cos C = \frac{13^2 + 10.5^2 - 13.289^2}{2(13)(10.5)}$ **[M2]**

($M1$ for correct substitution into implicit cosine rule)

67.9 (67.91...) **[A1]**

(c) Area $ABD = \frac{1}{2} \times 7.5 \times 9.2 \times \sin(105°)$ (33.32...) **[M1]**

Area $BCD = \frac{1}{2} \times 13 \times 10.5 \times \sin(67.9°)$ (63.24...) **[M1]**

96.6 (96.5... to 96.6) **[A1]**

[Total Marks: 9]

26 (a) Uses cosine rule: $BD^2 = 8.5^2 + 11.2^2 - 2(8.5)(11.2)\cos 105°$ **[M1]**

$BD^2 = 246.96...$ **[A1]**

15.7 **[A1]**

(b) Uses sine rule: $\frac{\sin C}{15.715} = \frac{\sin 38°}{11.0}$ **[M1]**

Acute angle $C = 61.59...°$ **[M1]**

Obtuse angle $BCD = 180° - 61.59...° = 118.4°$ **[A1]**

(c) Area of triangle $ABD = \frac{1}{2} \times 8.5 \times 11.2 \times \sin 105°$ $(= 45.97...)$ **[M1]**

Angle $DBC = 180° - 38° - 118.41° = 23.59°$ **[B1]**

Area of triangle $BCD = \frac{1}{2} \times 11.0 \times 15.715 \times \sin 23.59°$ $(= 34.59...)$ **[M1]**

80.6 **[A1]**

[Total Marks: 10]

27 **(a)** Angle $BAL = 085° - 050° = 35°$ **[B1]**

$\frac{\sin(ALB)}{40} = \frac{\sin 35°}{35}$ **[M1]**

$\sin(ALB) = \frac{40 \sin 35°}{35} = 0.6555\ldots$ **[M1]**

Acute angle $= 40.96°$, Obtuse angle $ALB = 180° - 40.96° = 139.0$ **[A1]**

(b) Angle $ABL = 180° - 35° - 139.04° = 5.96°$ **[B1]**

$\frac{AL}{\sin 5.96°} = \frac{35}{\sin 35°}$ **[M1]**

6.33 **[A1]**

(c) Bearing of A from $B = 180° + 050° = 230°$ **[M1]**

Bearing of L from $B = 230°$ − Angle $ABL = 230° - 5.96°$ **[M1]**

224.0 **[A1]**

[Total Marks: 10]